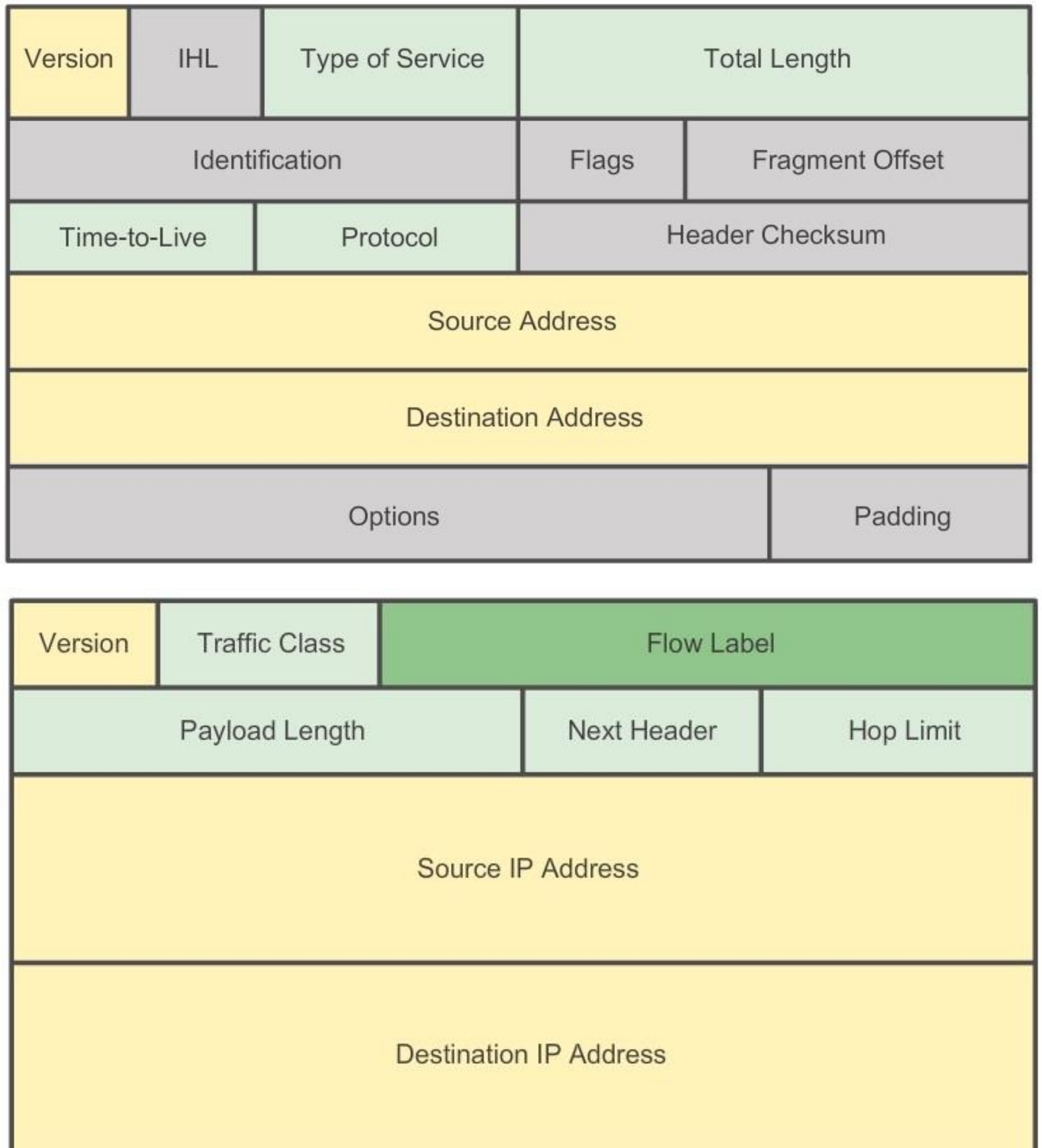
Comparison of IPv4-IPv6 headers



IPv6 addressing

Features:

- 128 bits long
- we write them as a series of hexadecimal values
- Consists of 32 hexadecimal digits
- we do not distinguish between lowercase and uppercase letters, you can use either

Representation of hexadecimal values

Representing Hexadecimal Values

Hexadecimal	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

Converting hexadecimal numbers

Hexadecimal Conversions of Binary Octets

Hexadecimal	Decimal	Binary
00	0	0000 0000
01	1	0000 0001
02	2	0000 0010
03	3	0000 0011
04	4	0000 0100
05	5	0000 0101
06	6	0000 0110
07	7	0000 0111
08	8	0000 1000
0A	10	0000 1010
0F	15	0000 1111
10	16	0001 0000
20	32	0010 0000
40	64	0100 0000
80	128	1000 0000
C0	192	1100 0000
CA	202	1100 1010
F0	240	1111 0000
FF	255	1111 1111

IPv6 address formats

- Preferred (primary) format
- Abbreviated (compact) formats

Preferred format examples

2001	:	0DB8	:	0000	:	1111	:	0000	:	0000	:	0000	:	0200
2001	:	0DB8	:	0000	:	00A3	:	ABCD	:	0000	:	0000	:	1234
2001	:	0DB8	:	000A	:	0001	:	0000	:	0000	:	0000	:	0100
2001	:	0DB8	:	AAAA	:	0001	:	0000	:	0000	:	0000	:	0200
FE80	:	0000	:	0000	:	0000	:	0123	:	4567	:	89AB	:	CDEF
FE80	:	0000	:	0000	:	0000	:	0000	:	0000	:	0000	:	0001
FF02	:	0000	:	0000	:	0000	:	0000	:	0000	:	0000	:	0001
FF02	:	0000	:	0000	:	0000	:	0000	:	0001	:	FF00	:	0200
0000	:	0000	:	0000	:	0000	:	0000	:	0000	:	0000	:	0001
0000	:	0000	:	0000	:	0000	:	0000	:	0000	:	0000	:	0000

IPv6 address abbreviation/compression rules

1. Leading zeros can be omitted from 16-bit parts or hextets . This rule only applies to leading zeros, not trailing zeros, otherwise the abbreviation would be ambiguous.

2. Any sequence of 16-bit segments (hextets) containing only zeros can be replaced with a double colon (::). The double colon can only be used once within an address, otherwise multiple possible address abbreviations would be the same.

Omitting leading zeros

Preferred	2001:0DB8:0000:1111:0000:0000:0000:0200
No leading 0s	2001: DB8: 0:1111: 0: 0: 0: 200

Using the double colon

Preferred	2001:0DB8:0000:1111:0000:0000:0000:0200
No leading 0s	2001: DB8: 0:1111: 0: 0: 0: 200
Compressed	2001:DB8:0:1111::200

Egy helytelen cím:

- 2001:0DB8::ABCD::1234

A helytelen cím többféleképpen is bővíthető:

- 2001:0DB8::ABCD:0000:0000:1234
- 2001:0DB8::ABCD:0000:0000:0000:1234
- 2001:0DB8:0000:ABCD::1234
- 2001:0DB8:0000:0000:ABCD::1234

IPv6 address types

- **Unicast - An IPv6 unicast** address uniquely identifies an interface on an IPv6-capable device.

- **Multicast - An IPv6 multicast** address is used to send a single IPv6 packet to multiple recipients.

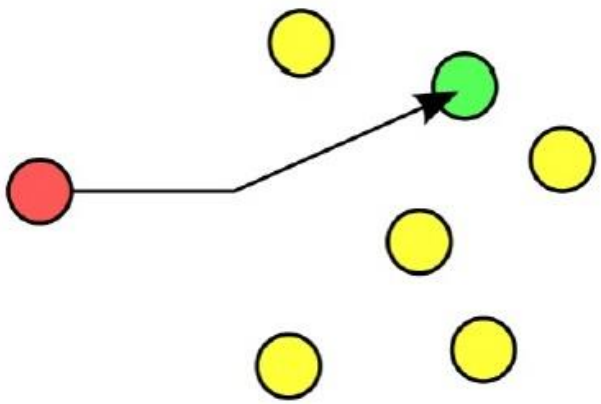
- **Anycast addressing -** An IPv6 anycast address is an IPv6 address that can be assigned to multiple devices. A packet sent to an anycast address is routed to the nearest device that has that address.

Comment:

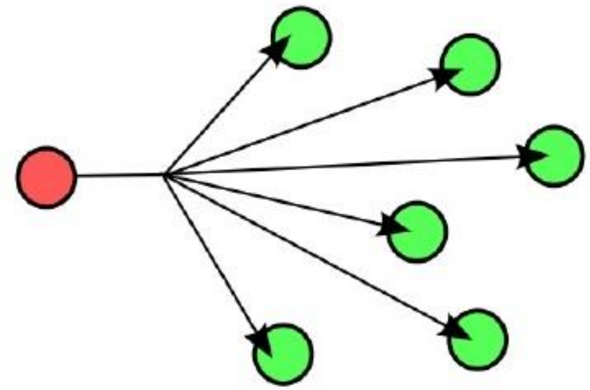
Unlike IPv4, IPv6 does not have broadcast addresses. However, there is an IPv6 all-node multicast address, which gives essentially the same result.

Addressing types

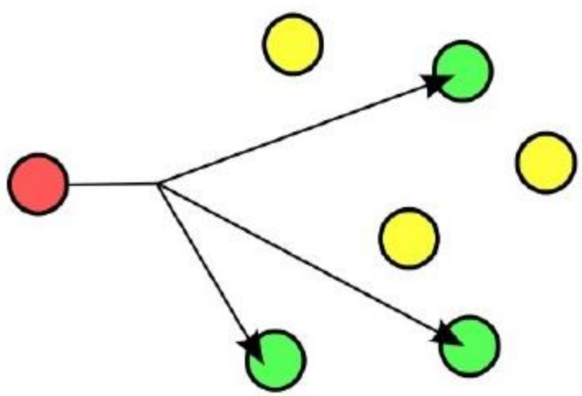
Unicast:



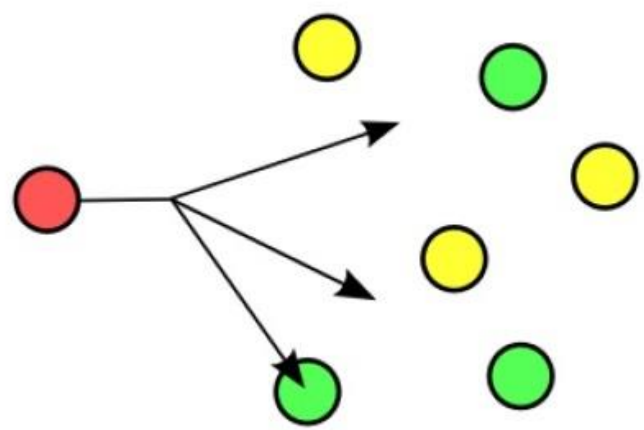
Broadcast:



Multicast:



Anycast:

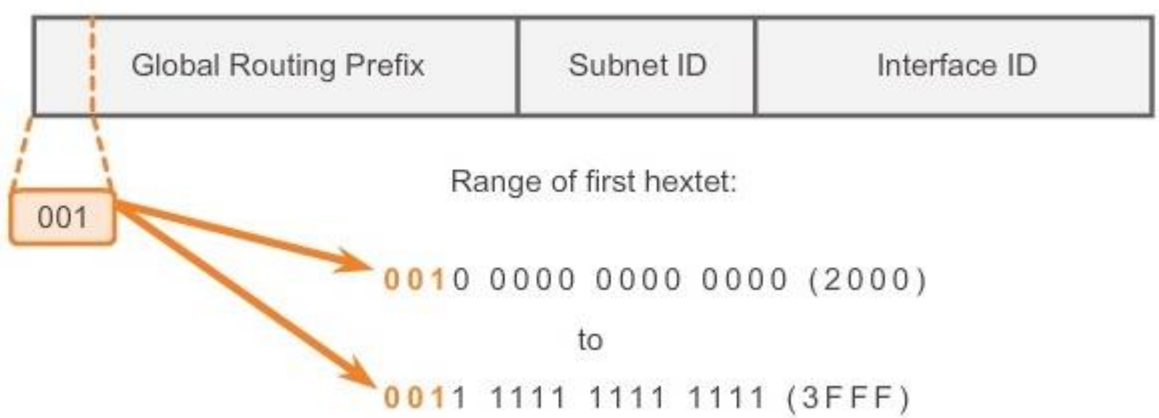


Unicast address types

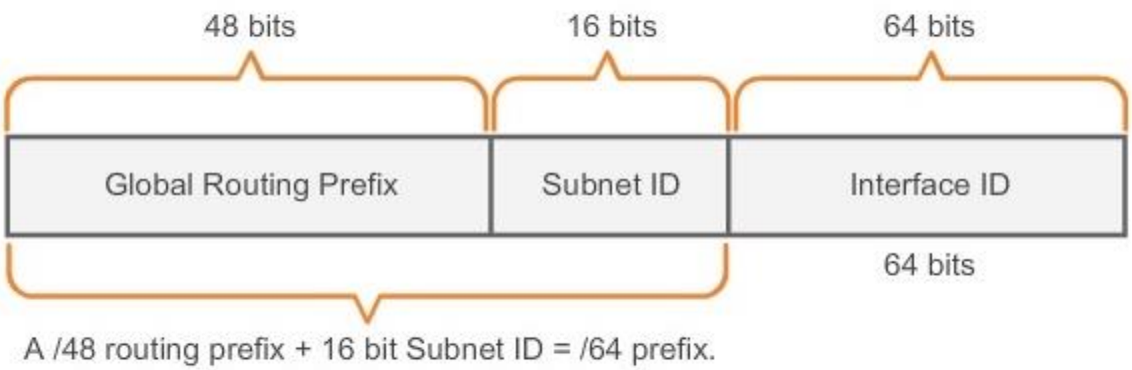
- Global unique address (global unicast)
- Link-local address
- Loopback address
- Unspecified address
- Unique local address
- Embedded IPv4 address

IPv6 Globally Unique Address

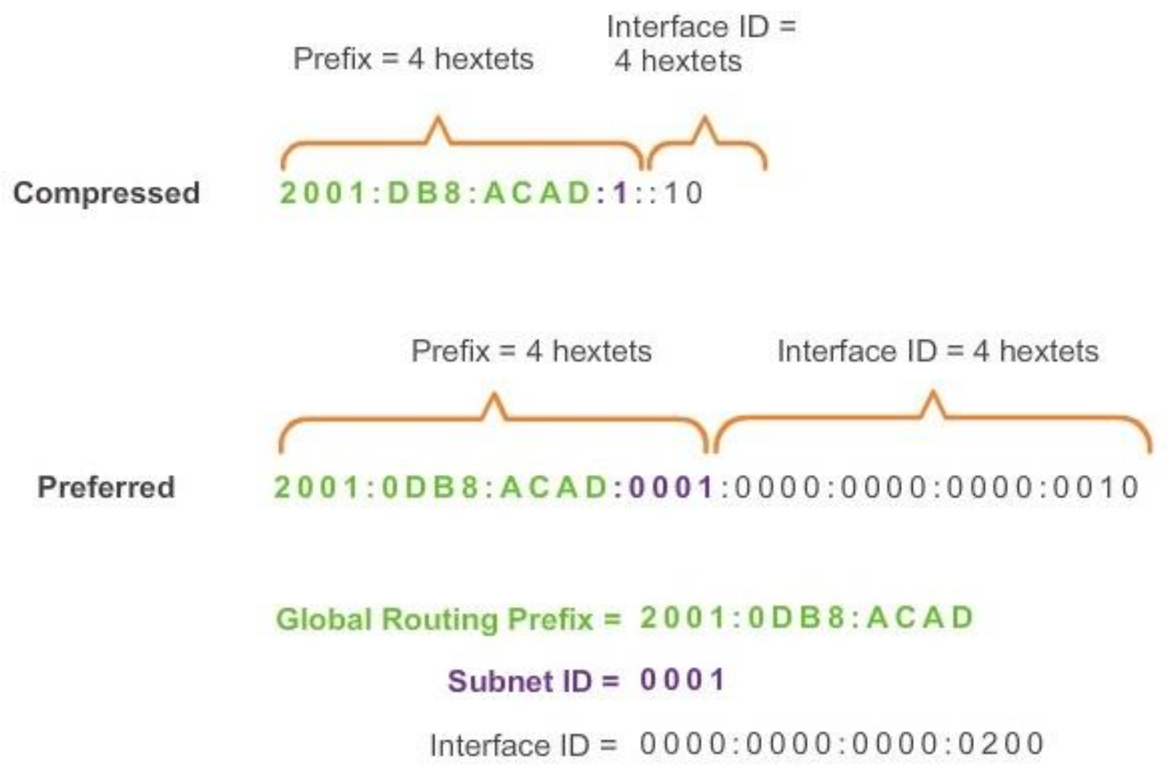
The portion of the IPv6 address space that is globally routable is reserved for unicast addressing. The allocation of addresses is done by IANA. Currently, prefixes within 2000::/3 are allocated . The structure of such addresses is shown in the figure below. The global routing prefix is the range assigned by IANA. The allocation of the subnet address range is similar to that known for IPv4. The interface ID is the address of the node's specific network interface. This can be generated manually, but automatic configuration is also possible.



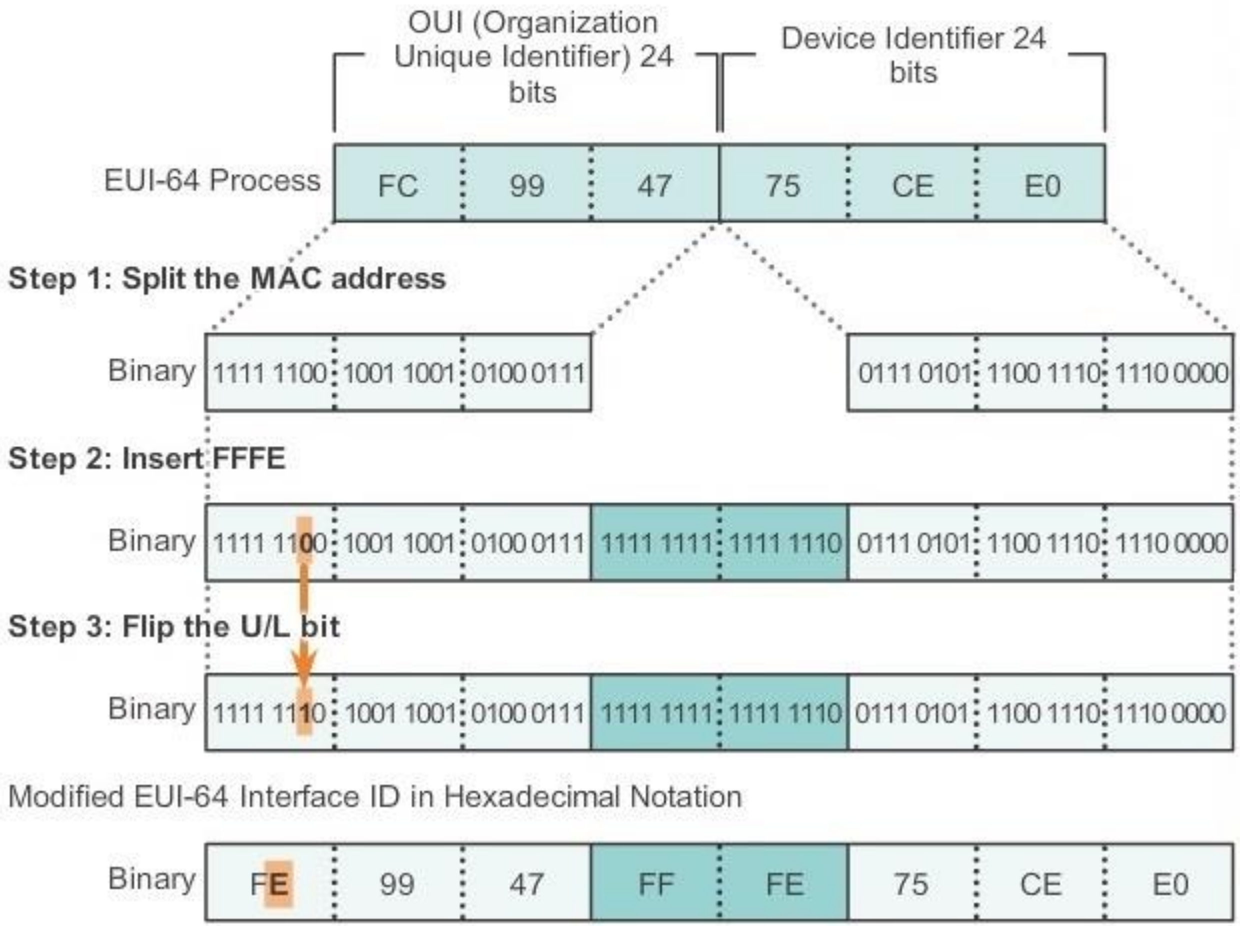
IPv6 /48 global routing prefix



Global unique address example



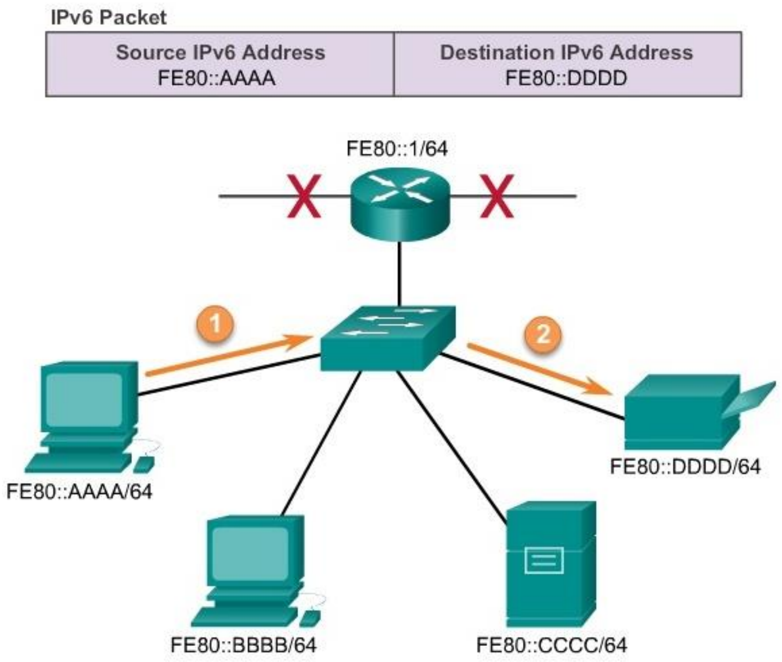
The EUI-64 method



Randomly generated interface ID

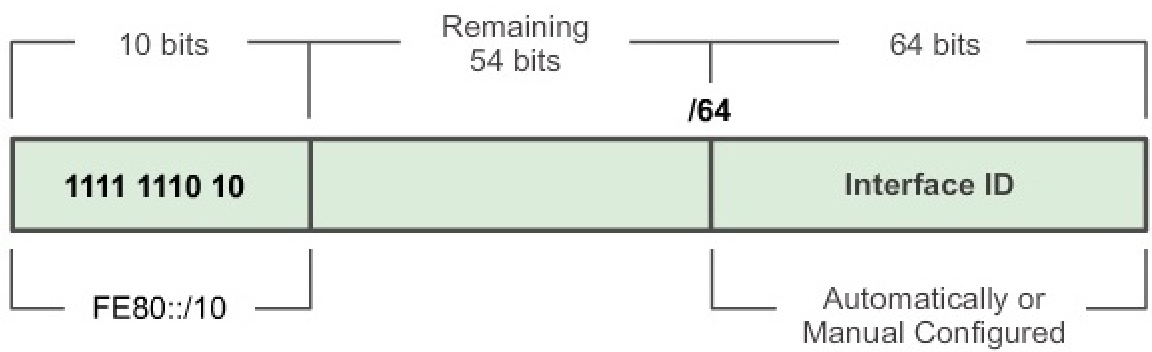
- Depending on the operating system, the device may use a randomly generated interface identifier instead of the MAC address and EUI-64 algorithm. For example, Windows operating systems starting with Vista use a randomly generated interface identifier instead of EUI-64. Windows XP and earlier Windows operating systems used EUI-64.

IPv6 link-local communication



Addresses that are valid only on a specific physical link, such as an Ethernet segment. IPv6 packets with such source addresses are not forwarded by routers . They are used in cases where, for example, a client device would like to communicate with it’s default gateway.

IPv6 link-local addressing



IPv6 loopback address

::1/128

A loopback address used similarly to the IPv4 loopback address **127.0.0.1** .

IPv6 undefined address

::/128

Unspecified Address , used when a node initializes itself.

IPv6 unique local addresses

FC00::/7

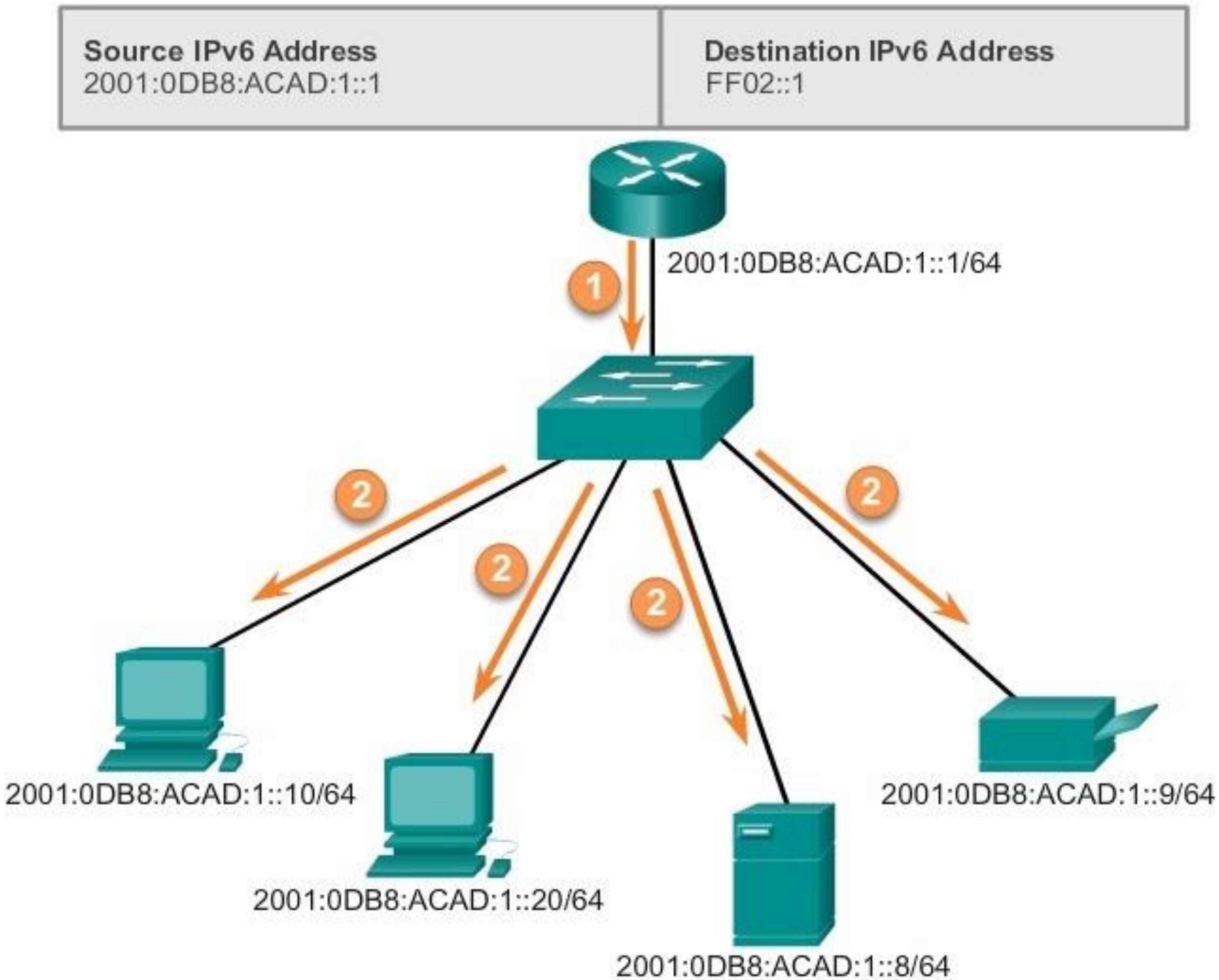
Unique Local Unicast Addresses (RFC 4193) By using such addresses, IPv6 can be used where there is no officially allocated IPv6 prefix , i.e. the purpose of their use is completely similar to private IPv4 addresses as defined in RFC 1918.

Embedded IPv4 addresses

::FFFF:/96

IPv4-Mapped IPv6 Address. Intended for use in IPv4 —> IPv6 transitions. An arbitrary **xyzw** IPv4 address is represented by the IPv6 address **::FFFF: xyzw** .

IPv6 multicast addressing



IPv6 multicast addressing

Unlike IPv4, there is no broadcast address in IPv6! The **FF00::/8 range can be used for** multicast addressing .

- **FF02::1** – *link local all nodes* – all devices on the given physical network (broadcast domain)
- **FF02::2** – *link local all routers* – all routers in the above circle
- **FF05::2** – *site local all routers* – the site routers

Comment:

The meaning of **FF02::1** is nothing more than the broadcast address for the current network in the case of IPv4!

IPv6 anycast addressing

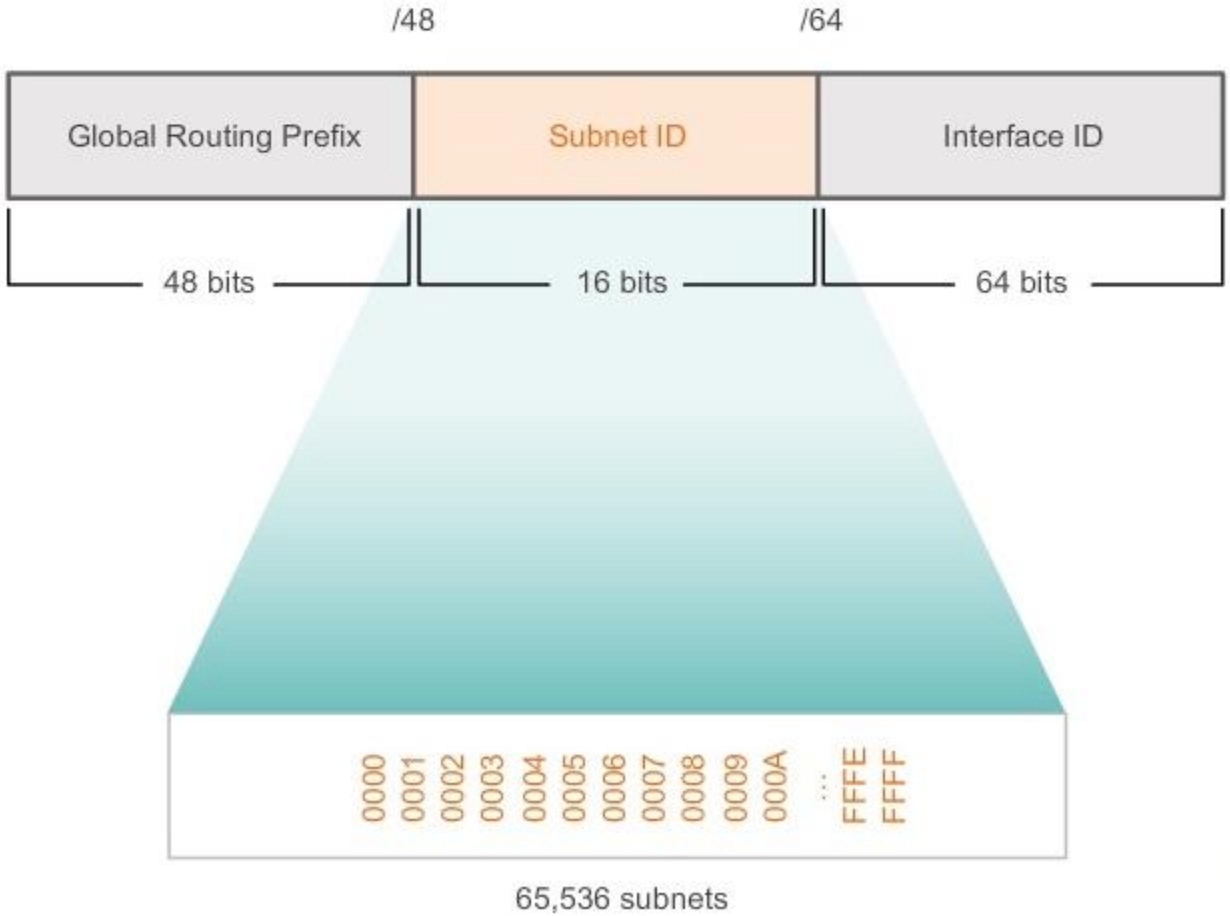
Anycast **addresses** are syntactically unicast addresses, but their usage is different. While in unicast each network interface has a different address, in anycast the same address is assigned to multiple network interfaces on different computers. A packet sent to an anycast address is delivered to the interface closest to the source (according to the metric of the routing protocol used in the network) with that anycast address.

The packet is addressed to one of the members of a group, but the network, not the sender, decides which one it should be. A typical choice is to be the group member closest to the sender.

IPv6 address types (summary)

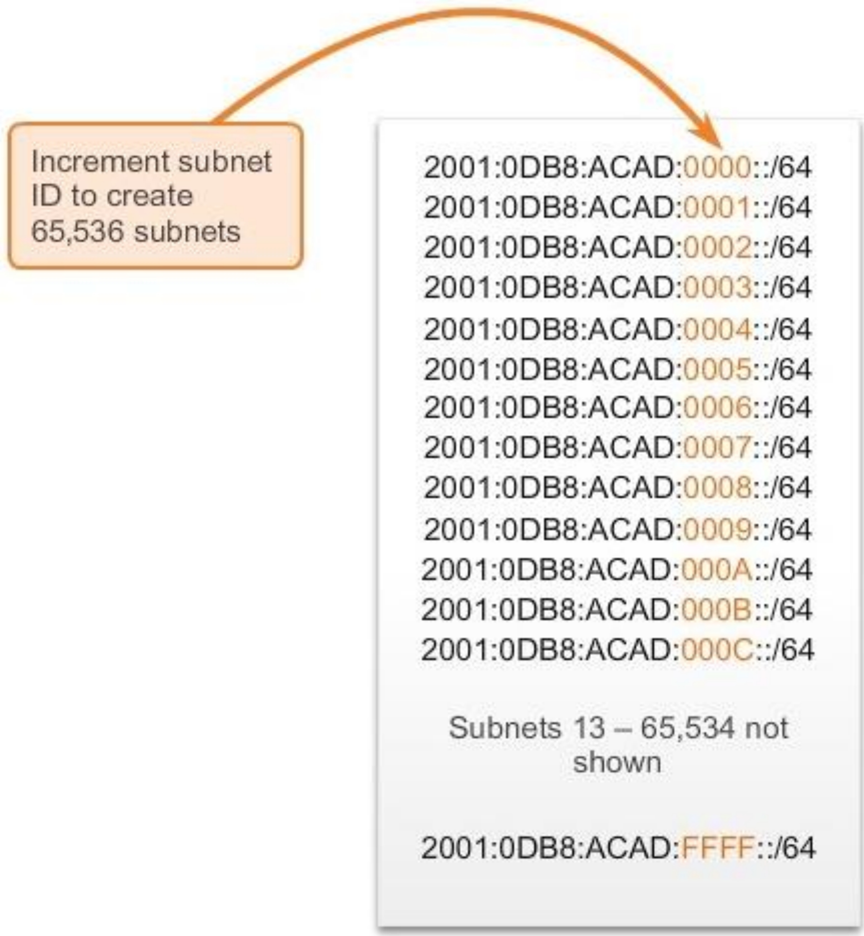
Address types		Identification sign
Reserved unicast 0000:/8	Unspecified	::/128
	Loopback	::1/128
Unicast	IPv4-Mapped IPv6	::FFFF:/96
	Link-Local	**FE80** ::/10
	Unique Local	FC00::/7 E.g. **FD**
	Global Unicast	The all other unallocated address ranges **2000** ::/3
Multicast		**FF** 00::/8

Creating subnets in an IPv6 network



Creating subnets in an IPv6 network

Address Block: 2001:0DB8:ACAD::/48



Querying the IPv6 address of a physical interface

- *ifconfig eth0*

- *ip addr show dev eth0*

Setting the IPv6 address of a physical interface

- *ifconfig eth0 inet6 add* 2001:ABCD::2/64
- *ip -6 addr add* 2001:ABCD::2/64 *dev eth0*

Routing table in IPv6

- *route –A inet6*

- *ip -6 route show*

Adding a new directly connected network in IPv6

- *route -A inet6 add* 2001:AAAA::/64 *dev eth0*

- *ip -6 route add* 2001:AAAA::/64 *dev eth0*

Setting a default gateway in IPv6

- *route -A inet6 add default gw* 2001: ABCD ::1

South

- *ip -6 route **add** default via* 2001: ABCD ::1

  *ip -6 route **del** default*

Ping command in IPv6

- *ping6* 2001:ABCD::2

Traceroute command in IPv6

- *traceroute6* 2001:ABCD::2